

Eleven: If $\alpha > 0$ be a number that satisfies $(-1)^\alpha = -1$. If we have $(X^2(t)) = t^\alpha + \pi + \sin(t)$ then $P_{XX} =$

- A) 0 B) 3π C) 2π D) π E) None

Q. Twelve: Which of the following is false?

- A) The null hypothesis is the one to be tested
B) In some cases both null and alternative hypothesis can be true.

☒ C) For $X \sim N(1, 2)$ then $P(X > 1) = 0.5$

D) Type I error occurs when we reject H_0 when it is actually true

Q. Thirteen: Consider the joint pdf $f(x, y) = x$ when $0 < x < 2$, $0 < y < b$ and zero otherwise. Then $f_X(x) =$

- ☒ A) $0.5x$ B) x C) $2y$ D) 2 E) None

Q. Fourteen ☒ X: For any constant $x > 0$, we have $u(x) = \delta(x)$

Q. Fifteen ☒ X: A correlation coefficient 0.97 indicates a strong positive correlation

Q. Sixteen ☒ X: The process $X(t) = A \cos(\omega t)$, $A \sim U(1, 2)$ is not first order stationary

Q. Seventeen ☒ X: The process $X(t) = \cos(\omega t)$, $\omega \sim U(1, 2)$ is first order stationary

Q. Eighteen ☒ X: For some independent random variables X, Y with $E(XY) = \bar{Y}$, they must have $\bar{X} = 1$ or $\bar{Y} = 0$

Q. Nineteen ☒ X: For $X \sim B(n, p)$ if $\frac{\text{Var}(X)}{E(X)} = 0.2$, then $p = 0.8$

Q. Twenty ☒ X: If $X \sim B(n, 0.5)$ and $E(X^2) = 5$, then $n = 5$

Q. One: For the process $X(t) = \cos(t + \theta)$ where $\theta \sim U(0, \pi)$, we have $E(X^2(t)) =$

A) 0.5π B) π C) 2π D) 3π E) None

Q. Two: The variance of a mean-ergodic process $X(t)$ with no periodic component and $R_{XX}(\tau) = \pi^2 + \pi e^{-|\tau|}$ is equal to

A) π^2 B) π C) 2π D) 3π E) None

Q. Three: For a random process $X(t) = A \cos(t)$ where $A \sim U(0, \pi)$ we have $P_{XX} =$

A) $\frac{1}{12}$ B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{1}{6}$ E) None

Q. Four: For $X \sim \text{Poisson}(m)$ with $P(X=0) = \sqrt{\frac{1}{e}}$; then $P(X > \bar{X}) =$

A) $\sqrt{\frac{1}{e}}$ B) $1 - \sqrt{\frac{1}{e}}$ C) $\frac{1}{e}$ D) $1 - \frac{1}{e}$ E) None

Q. Five: Jointly wide-sense stationary orthogonal processes $X(t), Y(t)$ with $R_{XY}(\tau) = 2 + \bar{X} \bar{Y}$, then $C_{XY}(t, t + \tau) =$

A) 1 B) 0 C) 2 D) 3 E) None

Q. Six: For jointly wide-sense stationary uncorrelated discrete time random processes $X(nT_s), Y(nT_s)$ such that $C_{XY}(nT_s, nT_s + kT_s) = 1 - \bar{X} \bar{Y}$. Then $R_{XY}(kT_s) + \bar{X} \bar{Y} =$

A) 2 B) 4 C) 6 D) 8 E) None

Q. Seven: If $\text{Var}(X) = 2$, and $Y = X - \bar{X}$, then the correlation of X and Y is

A) 1 B) 2 C) 3 D) 4 E) None

Q. Eight: The pdf of $Y = X + 1$; where $X \sim U(0, 1)$, equals

A) 2 B) 0.25 C) 0.5 D) 1 E) None

Q. Nine: The variance of the process $X(t) = A$ where $A \sim U(-\pi, \pi)$ is equal to

A) $\frac{\pi^2}{3}$ B) $\frac{\pi^3}{3}$ C) $\frac{2\pi^3}{3}$ D) $\frac{2\pi^2}{3}$ E) None

Q. Ten: For $X \sim N(a, a)$ if $\text{std}(X) = E(X)$, then $a + \text{std}(X) =$

A) 1 B) 2 C) 6 D) 4 E) None